Example 4.32 In how many ways 5 boys and 4 girls can be seated in a row so that no two girls are together.

Solution:

$$_B_1_B_2_B_3_B_4_B_5_$$

The 5 boys can be seated in the row in ${}^5P_5 = 5!$ ways. In each of these arrangements 6 gaps are created. Since no two girls are to sit together, we may arrange 4 girls in this 6 gaps. This can be done in ${}^6P_4$ ways. Hence, the total number of seating arrangements is

$$5! \times {}^6P_4 = 120 \times 360 = 43200.$$

Example 4.33 4 boys and 4 girls form a line with the boys and girls alternating. Find the number of ways of making this line.

Solution:
4 boys can be arranged in a line in ${}^4P_4 = 4!$ ways. By keeping boys as first in each of these arrangements, 4 gaps are created. In these 4 gaps, 4 girls can be arranged in ${}^4P_4 = 4!$ ways.

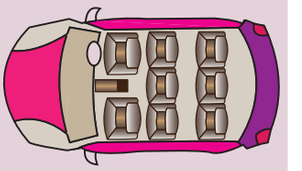
$$B_1_B_2_B_3_B_4_ \text{ or } G_1_G_2_G_3_G_4_$$

Therefore, keeping boys as first, the total number of arrangements are $4! \times 4!$. Similarly, keeping girls as first, by a similar argument, the total number of arrangements are $4! \times 4!$. Hence, by the rule of sum, keeping either a boy or a girl first, the total number of arrangements are

$$(4! \times 4!) + (4! \times 4!) = 2(4!)^2 = 1152.$$

Example 4.34 A van has 8 seats. It has two seats in the front with two rows of three seats behind. The van belongs to a family, consisting of seven members, $F, M, S_1, S_2, S_3,$ D_1, D_2. How many ways can the family sit in the van if

(i) There are no restriction?
(ii) Either F or M drives the van?
(iii) D_1, D_2 sits next to a window and F is driving?



Solution:

(i) As there 8 seats to be occupied out of which one seat is for the one who drives. Since there are no restrictions any one can drive the van. Hence the number of ways of occupying the driver seat is ${}^7P_1 = 7$ ways . The number of ways of occupying the remaining 7 seats by the remaining 6 people is ${}^7P_6 = 5040$. Hence the total number of ways the family can be seated in the car is $7 \times 5040 = 35280$.

(ii) As the driver seat can be occupied by only F or M, there are only two ways it can be occupied. Hence the total number of ways the family can be seated in the car is $2 \times 5040 = 10080$.

(iii) As there are only 5 window seats available for $D_1 \& D_2$ to occupy the number of ways of seated near the windows by the two family members is ${}^5P_2 = 20$. As the driver seat is occupied by F, the remaining 4 people can be seated in the available 5 seats in ${}^5P_4 = 120$. Hence the total number of ways the family can be seated in the car is $20 \times 1 \times 120 = 2400$.